

END SEMESTER EXAMINATIONS, SPRING 2025

Program: BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING), BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & BUSINESS SYSTEMS)/ BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING) (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)/ BACHELOR OF TECHNOLOGY (COMPUTER SCIENCE & ENGINEERING) (INTERNET OF THINGS, CYBERSECURITY AND BLOCKCHAIN TECHNOLOGY) / BACHELOR OF SCIENCE (HONOURS) (MATHEMATICS)	
Subject Name: MACHINE LEARNING/ MACHINE LEARNING ALGORITHM	
Subject Code: 1000015101/ 1000025101/ 1000105101/ 1000115101/ 1210015301	Semester: VI
Duration: 2 Hours	Maximum Marks: 50

Note: Answer the following questions as directed. Non-Programmable calculator allowed.

Mod. No.- Module/Topic/Unit No., **BL-** Blooms Taxonomy, **CO-** Course Outcome.

Group A

(Answer All questions)

(Each question carries 1 mark)

1. Write the correct answer from the options given: (10 X 1)

		Mod. No.	BL	CO
(i)	Which of the following options is/are true for K-fold cross-validation? 1. Increase in K will result in higher time required to cross validate the result. 2. Higher values of K will result in higher confidence on the cross-validation result as compared to lower values of K. 3. If K=N, then it is called Leave one out cross validation, where N is the number of observations.	IV	2	5
	(a) Only 1 (c) 2 and 3	(b) 1 and 2 (d) 1, 2 and 3		
(ii)	Which of the following libraries is used for boosting generalized additive models?	III	3	5
	(a) gamBoost (c) ada	(b) gbm (d) All of these		

		Mod. No.	BL	CO
(iii)	Which of the following statements about model complexity is TRUE?	IV	4	6
	(a) Higher model complexity leads to a lower chance of overfitting.			
	(b) Higher model complexity leads to a higher chance of overfitting.			
	(c) Reducing the number of features while adding feature interactions leads to a lower chance of overfitting.			
	(d) Reducing the number of features while adding feature interactions leads to a higher chance of overfitting.			

(iv)	Classification Analysis is to	III	2	4
	(a) Predict numeric answers			
	(b) Predict nonnumeric answers			
	(c) Predict string answers			
	(d) None of these			

(v)	Let's say, you are using an activation function X in the hidden layers of the neural network. At a particular neuron for any given input, you get the output as "-0.0001". Which of the following activation functions could X represent?	III	4	5
	(a) ReLU			
	(b) Sigmoid			
	(c) tanh			
	(d) All of these			

(vi)	One of the most common uses of Machine Learning today is in the domain of Robotics. Robotic tasks include a multitude of ML methods tailored towards navigation, robotic control and a number of other tasks. Robotic control includes controlling the actuators available to the robotic system. An example of this is control of a painting arm in automotive industries. The robotic arm must be able to paint every corner in the automotive parts while minimizing the quantity of paint wasted in the process. Which of the following learning paradigms would you select for training such a robotic arm?	I	4	3
	(a) Supervised Learning			
	(b) Unsupervised Learning			
	(c) Reinforcement Learning			
	(d) Semi-Supervised Learning			

(vii)	The effectiveness of an SVM depends upon:	III	1	3
	(a) Selection of the Kernel			
	(b) Kernel Parameters			
	(c) Soft Margin Parameter C			
	(d) All of these			

		Mod. No.	BL	CO
(viii)	Imagine, you are solving a classification problem with a highly imbalanced class. The majority class is observed 98% of times in the training data. Your model has 98% accuracy after taking the predictions on test data. Which of the following is true in such a case? 1. Accuracy metric is not a good idea for imbalanced class problems. 2. Accuracy metric is a good idea for imbalanced class problems. 3. Precision and recall metrics are good for imbalanced class problems. 4. Precision and recall metrics aren't good for imbalanced class problems.	IV	6	6
	(a) 1, 2, and 3 (b) 2 and 4 (c) 2, 3 and 4 (d) 1 and 3			
(ix)	Which one of the 3 Regularization techniques: Ridge, Lasso and Elastic Net, performs the fastest under the hood?	III	2	5
	(a) Ridge (b) Elastic Net (c) Lasso (d) None of the above			
(x)	Deep Learning uses deep Neural Networks for all these uses, except	VI	2	2
	(a) As an alternative to manual feature engineering (b) To uncover usually unobserved relationships in the data (c) Cases in which explain ability is the main objective (d) As a classification and regression technique			

Group B

(Answer any Four questions out of Six)
(Each question carries 5 marks)

(4 X 5)

2. (a) State the difference between supervised learning and unsupervised learning.
- (b) Compare and contrast linear regression and logistic regression. (3 + 2)

Mod. No.	BL	CO
I	1	3
III	1	1

	Mod. No.	BL	CO
3. (a) You are building a spam detection system for emails. You trained a model with high accuracy (98%), but users report that many spam emails are still reaching their inbox. What could be the problem?	III	4	4
(b) How would you evaluate and improve the model? (2 + 3)	IV	5	6
4. Consider two machine learning models: M1 and M2. It is given that model M1 is suffering from low bias and model M2 is suffering from high variance. How to tackle these situations and which algorithm should you use to tackle it for the same dataset? Justify your answer.	IV	6	4
5. Among 2000 people you are predicting 400 people to have cancer. In actuality only 300 people have the cancer and among those 300 people you have diagnosed 240 people correctly. Calculate the precision, specificity, recall, accuracy, and F1-scores in terms of confusion matrix.	IV	5	6
6. (a) Discuss the significance of the dimensionality reduction techniques.	V	2	2
(b) State two applications of Principal Component Analysis. (3 + 2)	V	3	2
7. (a) Explain the concept of vanishing and exploding gradients in deep neural networks.	VI	4	4
(b) How do they affect training, and what techniques can be used to mitigate these issues? (2 + 3)	VI	6	5

Group C

(Answer any Two questions out of Four)
(Each question carries 10 marks)

(2 X 10)

8. Cluster the following eight points into three clusters using the K-means clustering algorithm. The points are P1(2,10), P2(2,5), P3(8,4), P4(5,8), P5(7,5), P6(6,4), P7(1,2) and P8 (4,9). Consider cluster centers as C1 (2,10), C2 (5,8) and C3 (1,2).

9. Identify the first splitting attribute for the decision tree using information gain. The dataset is given below.

Major	Experience	Grade	Hired?
CS	Programming	A	NO
CS	Programming	A	NO
CS	Management	A	YES
CS	Management	B	YES
BUSINESS	Programming	A	YES
BUSINESS	Programming	B	YES
BUSINESS	Management	A	NO
BUSINESS	Management	A	NO

10. Consider the training data in the following table where Play is a class attribute.

Humidity	Sunny	Wind	Play
Low	No	Strong	No
High	No	Weak	Yes
High	Yes	Strong	Yes
High	No	Weak	Yes
Low	Yes	Strong	No

According to naive bayes classification, What is the class label for the following day (Humidity=Low, Sunny=No, Wind= Weak).

Mod. No.	BL	CO
V	6	4
II	3	4
III	3	4

Mod. No.	BL	CO
III	6	I

11. The grades of a class of 9 students on a midterm exam (X) and on the final exam (Y) are as follows:

X	77	50	71	72	81	94	96	99	67
Y	82	66	78	34	47	85	99	99	68

Find the linear regression hypothesis line and estimate the final examination grade of a student who received a grade of 85 on the midterm report.